

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – CODING CHALLENGE QUESTIONS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Coding challenge questions
Weightage:	40%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
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Section / Batch:	2025	Date:	28-07-2026

Instructions to Students

- Analyze the given scenario and identify the problem requirements before developing the solution.
- Design the algorithm or flowchart and select the appropriate 8085 instructions, registers, and addressing modes required for implementation.
- Develop and execute the 8085 Assembly Language Program (ALP) using a simulator or trainer kit to solve the given problem.
- Verify and validate the output by testing the program with the given input data and ensuring the expected results are obtained.
- Interpret the program execution by explaining the logic used, the role of registers, memory locations, flags, and the final output generated.

QUESTIONS

1. An electronics manufacturer stores inspection results of ten products as binary values:

- 00H – Defective
- FFH – Accepted

Before packaging, the controller must determine the number of accepted and defective products. Write an 8085 ALP to

- Count accepted products.
 - Count defective products.
 - Compare both counts and indicate whether production quality is satisfactory.
2. A weather station records hourly temperature values in memory. The controller must determine whether the readings show an increasing trend. Develop an 8085 ALP that compares consecutive readings and counts how many times the temperature increased relative to the previous reading.
 3. A memory block contains a mixture of even and odd data values collected from different sensors. For optimized processing, the controller must separate these values. Write an 8085 ALP to:
 - Count the number of even values.
 - Count the number of odd values.
 - Store the two counts in memory.

Code of Conduct

I V.Sriram(192524037) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: V.Sriram

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly understands problem constraints, edge cases, and competitive requirements	Understands problem with minor gaps in constraints	Partial understanding; misses key constraints	Misinterprets problem or constraints
Algorithm	25%	Selects optimal algorithm with best time and space complexity for competitive constraints	Uses efficient algorithm with minor scope for improvement	Uses basic/inefficient approach	No clear algorithmic approach
Code Implementation	20%	Writes correct, efficient, and clean code that passes all constraints	Code works with minor errors or inefficiencies	Partially correct code; fails for some cases	Code does not execute or gives incorrect results
Competitive Performance	20%	Solves problem within time limits and handles large inputs effectively	Solves within acceptable time with minor delays	Struggles with time limits or larger inputs	Unable to solve within time constraints
Test Case Handling	15%	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases

Marks Evaluation:

Q. No	Problem Understanding (20)	Algorithm (25)	Code Implementation (20)	Competitive Performance (20)	Test case handling (15)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 25	/ 20	/ 20	/ 15	/ 100

Course Coordinator

Faculty Incharge

GNUSim8085 - 8085 Microprocessor Simulator

FileResetAssemblerDebugHelp

Registers

A01

B00

C06

D04

E00

H80

L5A

Flags

S0

Z0

AC0

P0

CY0

Hexadecimal

0

→ To Hex

← To Dec

I/O Ports

00-00

Update Port Value

Memory

00-FF

Update Memory

1; COUNT ACCEPTED AND DEFECTIVE PRODUCTS

2; 00H = DEFECTIVE, FFH = ACCEPTED

3LXI H,8050H

4MVI B,0AH

5MVI C,00H

6MVI D,00H

7

8LOOP: MOV A,M

9CPI 0FFH

10JZ ACCEPT

11

12INR D

13JMP NEXT

14

15ACCEPT: INR C

16

17NEXT: INX H

18DCR B

19JNZ LOOP

20

21MOV A,C

22STA 8060H

23

24MOV A,D

25STA 8061H

26

27MOV A,C

28CMP D

29

30JC NOTGOOD

31

32MVI A,01H

33STA 8062H

34HLT

35

36NOTGOOD: MVI A,00H

37STA 8062H

38HLT

DataStackKeyPadMemoryI/O Ports

Start8050OK

Address (Hex)AddressData

1F728050FF

1F73805100

1F748052FF

1F758053FF

1F76805400

1F778055FF

1F78805600

1F798057FF

1F7A8058FF

1F7B805900

Line NoAssembler Message

100000LXI H,8050H

200003MVI B,0AH

300005MVI C,00H

400007MVI D,00H

500009MOV A,M

.

.

.

380003BHLT

Assembly Successful

No Errors

Simulator Status : Program executed successfully.

GNUSim8085 - 8085 Microprocessor Simulator

FileResetAssemblerDebugHelp

Registers

A05

B00

C05

D05

E00

H80

L5A

Flags

S0

Z1

AC0

P1

CY0

Hexadecimal

00

→ To Hex

→ To Dec

I/O Ports

00

-

+

00

Update Port Value

Memory

00

-

+

FF

Update Memory

1LXI H, 8050H

2MVI B, 0AH

3MVI C, 00H

4MVI D, 00H

5

6LOOP: MOV A, M

7ANI 01H

8JZ EVEN

9INR D

10JMP NEXT

11

12EVEN: INR C

13

14NEXT: INX H

15DCR B

16JNZ LOOP

17

18MOV A, C

19STA 8060H

20MOV A, D

STA 8061H

HLT

DataStackKeyPadMemoryI/O Ports

Start8050OK

Address (Hex)	Address	Data
1F72	8050	25
1F73	8051	42
1F74	8052	17
1F75	8053	50
1F76	8054	33
1F77	8055	44
1F78	8056	11
1F79	8057	26
1F7A	8058	39
1F7B	8059	18

Line No	Assembler Message
1	0000 LXI H,8050H
2	0003 MVI B,0AH
3	0005 MVI C,00H
4	0007 MVI D,00H
.	
22	0024 HLT

Assembly Successful
No Errors

Simulator Status : Program executed successfully.

GNUSim8085 - 8085 Microprocessor Simulator

FileResetAssemblerDebugHelp

Registers

A49

B00

C05

D00

E00

H80

L55

Flags

S0

Z0

AC0

P1

CY0

Hexadecimal

00

→ To Hex

→ To Dec

I/O Ports

00-00

Update Port Value

Memory

00-FF

Update Memory

1LXI H,8050H

2MOV A,M

3MVI B,04H

4

5LOOP: INX H

6CMP M

7JNC NEXT

8MOV A,M

9

10NEXT: DCR B

11JNZ LOOP

12

13STA 8060H

14HLT

DataStackKeyPadMemoryI/O Ports

Start8050OK

Address (Hex)AddressData

1F7280502A

1F73805115

1F7480523C

1F75805308

1F76805449

Line NoAssembler Message

10000LXI H,8050H

20003MOV A,M

30004MVI B,04H

40006INX H

50007CMP M

60008JNC 000A

7000AMOV A,M

8000BDCCR B

9000CJNZ 0006

10000E STA 8060H

110011HLT

Assembly Successful

No Errors

Simulator Status : Program executed successfully.